The documentation and process conversion measures necessary to comply with this revision shall be completed by 25 December 1994

INCH POUND

MIL-S-19500/370C
25 September 1994
SUPERSEDING
MIL-S-19500/370B
26 February 1990

MILITARY SPECIFICATION

SEMICONDUCTOR DEVICE, TRANSISTOR, NPN, SILICON, HIGH-POWER
TYPE 2N3442, JAN, JANTX AND JANTXV

This Specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for NPN silicon, high-power transistors. Two levels of product assurance are provided for each device type as specified in MIL-S-19500.

1.2 Physical dimensions. See figure 1.

1.3 Maximum ratings.

P_T 1/ T_C = +25°C	P_T 2/ T_C = +25°C	V_{CBO}	V_{CEO}	V_{EBO}	V_{CER}	I_B	I_C	T_{OP} and T_{STG}
W	W	V dc	V dc	V dc	V dc	A dc	A dc	°C
6.0	117	160	140	7	150	7	10	-55 to +200

1/ Derate linearly 34.2 mW/°C above T_A = +25°C.
2/ Derate linearly 668 mW/°C above T_C = +25°C.

1.4 Primary electrical characteristics.

	h_{FE1} 1/	$V_{CE(sat)}$ 1/	h_{fe}	$R_{\Theta JC}$
	V_{CE} = 4 V dc I_C = 3 A dc	I_C = 3 A dc I_B = 300 mA dc	V_{CE} = 4 V dc I_C = 3 A dc f = 100 kHz	
		V dc		°C/W
Min	20		1.0	
Max	70	1.0		1.5

1/ Pulsed (see 4.5.1).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Electronics Supply Center, ATTN: DESC-ELDT, 1507 Wilmington Pike, Dayton, OH 45444-5765 by using the Standardization Document Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A

FSC 5961

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

MILITARY

MIL-S-19500 - Semiconductor Devices, General Specification for.

STANDARD

MILITARY

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Associated detail specification. The individual item requirements shall be in accordance with MIL-S-19500 and as specified herein.

3.2 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-S-19500.

3.3 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-S-19500, and on figure 1 herein. Current density of internal conductors shall be specified in 3.6.5 of MIL-S-19500.

3.3.1 Lead material and finish. Lead material shall be Kovar or Alloy 52. Lead finish shall be solderable in accordance with MIL-S-19500, MIL-STD-750, and herein. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

3.4 Marking. Marking shall be in accordance with MIL-S-19500.

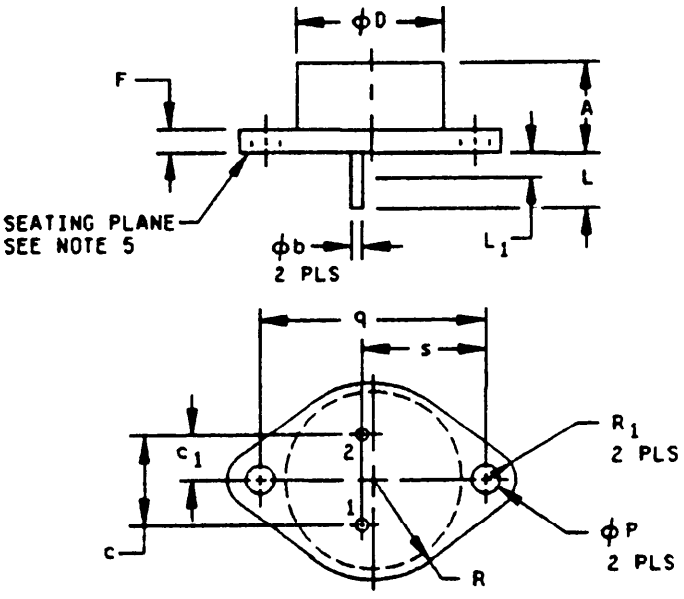
4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-S-19500, and as specified herein.

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-S-19500.

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Ltr	Dimensions				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
A	.270	.350	6.86	8.89	
ϕb	.038	.043	0.97	1.09	
c	.420	.440	10.67	11.18	4,5
c_1	.205	.225	5.21	5.72	4,5
ϕD		.875		22.22	
F	.060	.135	1.52	3.43	
L	.312	.500	7.92	12.70	
L_1		.050		1.27	
ϕP	.151	.165	3.84	4.19	
q	1.177	1.197	29.90	30.40	
R	.495	.525	12.57	13.34	
R_1	.131	.188	3.33	4.78	
s	.655	.675	16.64	17.15	



NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Terminal 1, base; terminal 2, emitter; case, collector.
4. Measured at points .050 inch (1.27 mm) - .055 inch (1.4 mm) below the seating plane. When gauge is not used, measurement will be made at the seating plane.
5. The seating plane of the header shall be flat within .001 inch (0.03 mm) concave to .004 inch (0.10 mm) convex inside a .930 inch (23.62 mm) diameter circle on the center of the header and flat within .001 inch (0.03 mm) concave to .006 inch (0.15 mm) convex overall.

FIGURE 1. Physical dimensions.

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4.3 Screening (JANTX and JANTXV levels). Screening shall be in accordance with table II of MIL-S-19500, and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see table II of MIL-S-19500)	Measurement JANTX, JANTXV levels only
11	h_{FE1}, I_{CEX}
12	See 4.3.1
13	Subgroup 2 of table I herein. ΔI_{CEX} = 100% or 500 μA dc whichever is greater; Δh_{FE1} = 25% of initial value.

4.3.1 Power burn-in conditions. Power burn-in conditions are as follows:

T_J = +187.5°C ±12.5°C; V_{CB} = 24 V dc; $T_A \leq$ +100°C.

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-S-19500.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-S-19500 and table I herein. Electrical measurements (end-points) and delta requirements shall be in accordance with the applicable steps of table II herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table IVb (JANTX and JANTXV) of MIL-S-19500. Electrical measurements (end-points) and delta requirements shall be in accordance with the applicable steps of table II herein.

4.4.2.1 Group B inspection, table IVb (JANTX and JANTXV) of MIL-S-19500.

Subgroup	Method	Conditions
3	1037	ΔT_J > +100°C for 2,000 cycles; V_{CB} > 24 V dc.
5	3131	See 4.5.2.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table V of MIL-S-19500. Electrical measurements (end-points) and delta requirements shall be in accordance with the applicable steps of table I herein.

4.4.3.1. Group C inspection, table V of MIL-S-19500.

Subgroup	Method	Conditions
2	2036	Test condition A, weight = 10 lbs, ±10 oz., t = 15 s.
6	1037	ΔT_J > +100°C for 6,000 cycles; V_{CB} > 24 V dc.

4.5 Method of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

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TABLE I. Group A inspection. 1/

Inspection	MIL-STD-750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroup 1						
Visual and mechanical examination	2071					
Subgroup 2						
Collector to emitter breakdown voltage	3011	Bias condition D; I_C = 3 A dc; Pulsed (see 4.5.1)	$V_{(BR)CEO}$	140		V dc
Collector to emitter breakdown voltage	3011	Bias condition B; I_C = 1.5 A dc; R_{BE} = 100Ω; See figure 2	$V_{(BR)CER}$	150		V dc
Collector to emitter breakdown voltage	3011	Bias condition A; I_C = 1.5 A dc; V_{EB} = 1.5 V dc; See figure 2	$V_{(BR)CEX}$	160		V dc
Collector to emitter cutoff current	3041	Bias condition A; V_{CE} = 140 V dc; V_{EB} = 1.5 V dc	I_{CEX}		1.0	mA dc
Emitter to base cutoff current	3061	Bias condition D; V_{EB} = 7 V dc	I_{EBO}		1.0	mA dc
Forward-current transfer ratio	3076	V_{CE} = 4 V dc; I_C = 3 A dc; Pulsed (see 4.5.1)	h_{FE1}	20	70	
Saturation voltage and resistance	3071	I_C = 3 A dc; I_B = 300 mA dc; Pulsed (see 4.5.1)	$V_{CE(sat)}$		1.0	V dc
Base emitter (nonsaturated) voltage	3066	Test condition B; V_{CE} = 4 V dc; I_C = 3 A dc; Pulsed (see 4.5.1)	V_{BE}		1.7	V dc

See footnote at end of table.

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TABLE I. Group A inspection 1/ - Continued.

Inspection	MIL-STD-750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroup 3						
High-temperature operation		T_A = +150°C				
Collector to base cutoff current	3036	Bias condition D; V_{CB} = 140 V dc	I_{CBO}		10	mA dc
Low-temperature operation		T_A = -55°C				
Forward-current transfer ratio	3076	V_{CE} = 4 V dc; I_C = 3 A dc; Pulsed (see 4.5.1)	h_{FE2}	15		
Subgroup 4						
Small-signal short-circuit forward-current transfer ratio	3306	V_{CE} = 4 V dc; I_C = 3 A dc; f = 100 kHz	h_{fe}	1.0		
Subgroup 5						
Safe operating area (continuous dc)	3051	T_C =+25°C; t = 1 s				
Test 1		V_{CE} = 11.7 V dc; I_C = 10 A dc				
Test 2		V_{CE} = 78 V dc; I_C = 1.5 A dc				
Test 2		V_{CE} = 140 V dc; I_C = 0.5 A dc; See figure 3				
Electrical measurements		See table I, subgroup 2 herein				
Subgroups 6 and 7						
Not applicable						

1/ For sampling plan, see MIL-S-19500.

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TABLE II. Groups A, B, and C electrical end-point measurements. 1/ 2/

Step	Inspection	MIL-STD-750		Symbol	Limits		Unit
		Method	Conditions		Min	Max	
1.	Collector to emitter cutoff current	3041	Bias condition A; V_{CE} = 140 V dc V_{EB} = 1.5 V dc	I_{CEX}		1.0	mA dc
2.	Collector to emitter cutoff current	3041	Bias condition A; V_{CE} = 140 V dc V_{EB} = 1.5 V dc	I_{CEX}		2.0	mA dc
3.	Emitter to base cutoff current	3061	Bias condition D; V_{EB} = 7 V dc	I_{EBO}		1.0	mA dc
4.	Emitter to base cutoff current	3061	Bias condition D; V_{EB} = 7 V dc	I_{EBO}		2.0	mA dc
5.	Forward current transfer ratio	3076	V_{CE} = 4 V dc; I_C = 3 A dc; pulsed (see 4.5.1)	h_{FE1}	20	70	
6.	Forward current transfer ratio	3076	V_{CE} = 4 V dc; I_C = 3 A dc; pulsed (see 4.5.1)	h_{FE1}	18	85	
7.	Collector to emitter cutoff current	3041	Bias condition A; V_{CE} = 140 V dc V_{EB} = 1.5 V dc	ΔI_{CEX} 3/		100 percent of initial reading or 500 µA dc, whichever is greater.	

1/ The electrical measurements for table IVb (JANTX and JANTXV) of MIL-S-19500 are as follows:

a. Subgroup 2, see table II herein, steps 1, 3 and 5.

b. Subgroup 3, see table II herein, steps 2, 4, 6 and 7.

c. Subgroup 6, see table II herein, steps 2, 4, 6 and 7.

2/ The electrical measurements for table V of MIL-S-19500 are as follows:

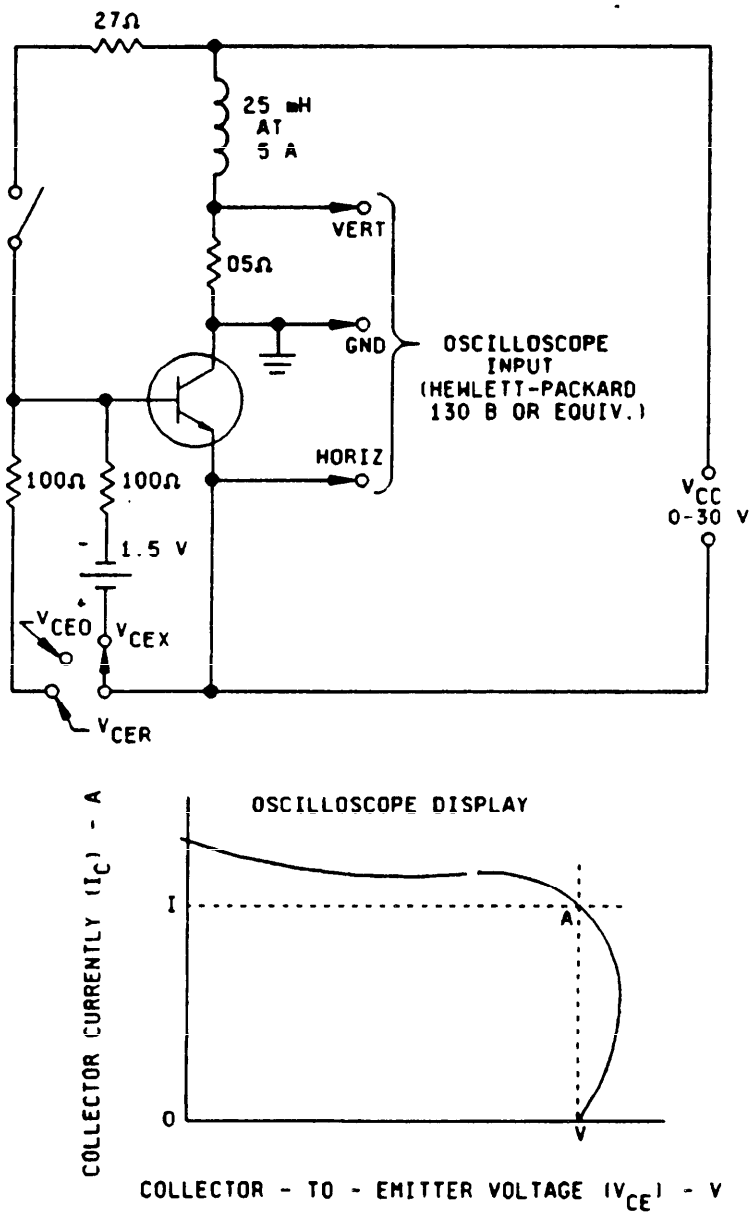
a. Subgroup 2, see table II herein, steps 1 and 5.

b. Subgroup 3, see table II herein, steps 1 and 5.

c. Subgroup 6, see table II herein, steps 2, 4, 6 and 7.

3/ Devices which exceed the group A limits for this test shall not be accepted.

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NOTE: The voltages V_{CEO}, V_{CER}, or V_{CEX} are acceptable when the trace falls to the right and above point "A".

FIGURE 2. Test circuit for V_{CEO}, V_{CER}, and V_{CEX}.

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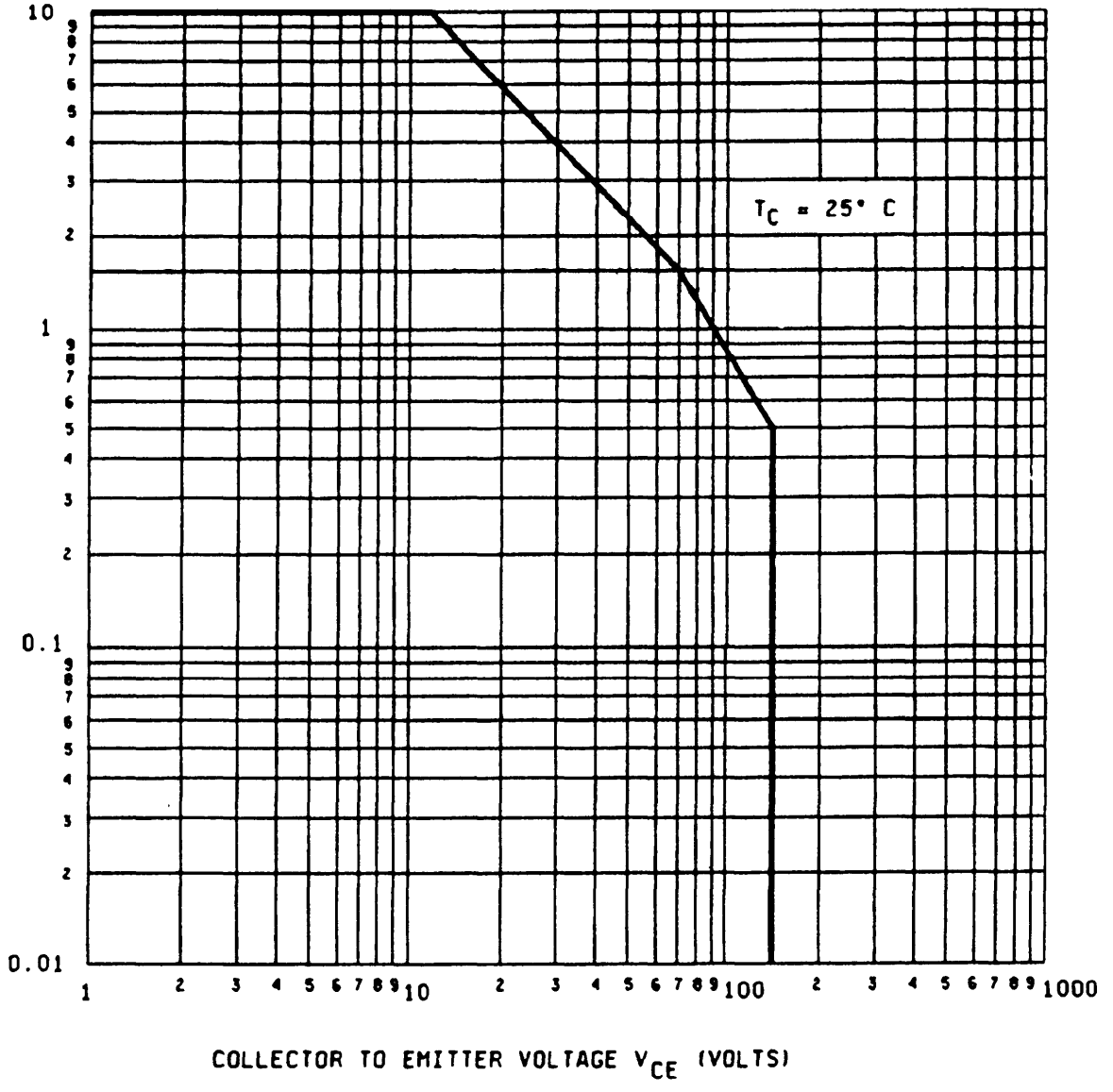


FIGURE 3. Maximum safe operating graph (dc).

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4.5.2 Thermal resistance. Thermal resistance measurements shall be conducted in accordance with test method 3131 of MIL-STD-750. The following details shall apply.

a. Collector current magnitude during power applications shall be 2 A dc.

b. Collector to emitter voltage magnitude shall be 20 V dc.

c. Reference temperature measuring point shall be the case.

d. Reference point temperature shall be +25°C ≤ T_R ≤ +75°C and recorded before the test is started.

e. Maximum limit for $R_{\Theta JC}$ shall be 1.5°C/W.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-S-19500.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Notes. The notes specified in MIL-S-19500 are applicable to this specification.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Issue of DODISS to be cited in the solicitation.

b. Lead finish as specified (see 3.3.1).

c. Type designation and quality assurance level.

6.3 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 17

Review activities:
Army - AR, MI
Navy - AS, CG, MC
Air Force - 19, 85, 99

Preparing activity:
DLA - ES

Agent:
DLA - ES

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INSTRUCTIONS

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3. The preparing activity must provide a reply within 30 days from receipt of the form.

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-S-19500/370C	2. DOCUMENT DATE (YYMMDD) 940925

3. DOCUMENT TITLE
SEMICONDUCTOR DEVICE, TRANSISTOR, NPN, SILICON, HIGH-POWER, TYPE 2N3442, JANTX AND JANTXV

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON
c. ADDRESS (Include Zip Code)	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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